Chapter 8: The binomial expansion

Chapter 8 · Original slide 1 · Binomial expansion

Chapter 8 · Pure Year 1

Chapter 8 · Original slide 2 · Starters

1) Expand and simplify where possible:

b) $(2x-3y)^2$

c) $(x-y)^3$

d) $(2+x)^3$

6) Simplify:

g) $(-2x)^3$ c) $\left(\frac{2}{5}x\right)^2$

h) $(3x)^{-4}$ d) $\left(\frac{1}{3}x\right)^{-3}$

10) Simplify using root notation:

k) $(25x)^{\frac{1}{2}}$ c) $\left(\frac{9}{100}x\right)^{-\frac{1}{2}}$

l) $(64x)^{-\frac{2}{3}}$ d) $\left(\frac{8}{27}x\right)^{\frac{4}{3}}$

Chapter 8: The binomial expansion

Chapter 8 · Original slide 3 · Chapter overview

1. Pascal's triangle

2. Factorial notation

3. Binomial expansion

4. Using expansions for estimation

Find the first four terms in $(4+5x)^{10}$, in ascending powers of x.

Use an expansion to estimate 1.05^{10} to five decimal places.

Given $\binom{8}{3} = \frac{8!}{3!a!}$, find a.

Chapter 8 · Original slide 4 · Patterns in binomial expansions

a)
Expand $(a+b)^0$.

b)
Expand $(a+b)^1$.

c)
Expand $(a+b)^2$.

d)
Expand $(a+b)^3$.

e)
Expand $(a+b)^4$.

What do you notice?
Describe the coefficients and the powers of a and b.

Chapter 8 · Original slide 5 · Pascal's triangle

Each entry, except the outside 1s, is the sum of the two entries immediately above it.

Memorise these first rows.

Chapter 8 · Original slide 6 · Expanding (2 + 3x)⁴

Find the expansion of $(2+3x)^4$.

1. Use row 4 of Pascal's triangle.

Chapter 8: The binomial expansion

Chapter 8 · Original slide 7 · Expanding a bracket with subtraction

Expand $(1-2x)^3$.

Write the bracket as $1+(-2x)$. The second term is $-2x$.

Chapter 8 · Original slide 8 · Finding a single coefficient

The coefficient of x^2 in $(2-cx)^5$ is 720. Find the possible values of c.

Chapter 8 · Original slide 9 · Test your understanding

(a) Find the first three terms, in ascending powers of x, of $(2 + kx)^7$, where k is a constant. Give each term in its simplest form. (4 marks)
Given that the coefficient of x^2 is six times the coefficient of x, (b) find k. (2 marks)

Chapter 8 · Original slide 10 · Exercise 8A · Questions 4 and 5

Question 4

4 Fully expand the expression $(1 + 3x)(1 + 2x)^3$.

Question 5

5 Expand $(2 + y)^3$. Hence or otherwise, write down the expansion of $(2 + x - x^2)^3$ in ascending powers of x.

Chapter 8 · Original slide 11 · Exercise 8A · Questions 6 and 7

Question 6

6 The coefficient of x^2 in the expansion of $(2+ax)^3$ is 54. Find the possible values of the constant a.

Question 7

7 The coefficient of x^3 in the expansion of $(2-x)(3+bx)^3$ is 45. Find possible values of the constant b.

Chapter 8 · Original slide 12 · Exercise 8A – Y1 page 160-161

8 Work out the coefficient of x^2 in the expansion of $(p-2x)^3$. Give your answer in terms of p.

9 After 5 years, the value of an investment of \pounds500 at an interest rate of $X\%$ per annum is given by

$$500\left(1+\frac{X}{100}\right)^5$$

Find an approximation for this expression in the form $A+BX+CX^2$, where A, B and C are constants to be found. You can ignore higher powers of X.

Chapter 8: The binomial expansion

Chapter 8 · Original slide 13 · Extension · A binomial equation

How many pairs of positive integers x, y satisfy $x^3 + 6x^2y + 12xy^2 + 8y^3 = 2^{30}$?

A) 0
B) 2^6
C) $2^9 - 1$
D) $2^{10} + 2$
E) — (blank in the supplied slide)

Chapter 8 · Original slide 14 · Factorial and the choose function

$n! = n(n-1)\cdots 2 \cdot 1$
Say “n factorial”. It counts the ways of arranging n distinct objects in a line.

$\binom{n}{r} = \frac{n!}{r!(n-r)!}$
Say “n choose r”. It counts selections of r objects from n, when order does not matter.

Three letters A, B and C have $3 \times 2 \times 1 = 3! = 6$ arrangements. Use the factorial button on your calculator.

Choosing four people from ten gives $\binom{10}{4} = \frac{10!}{4!6!} = 210$ selections. Use the nCr button. The $\binom{n}{r}$ notation is preferable in written work.

Chapter 8 · Original slide 15 · Factorial and choose examples

Calculate these values using the factorial button, without the nCr button.

a)
$5!$

b)
$\binom{5}{3}$

c)
$0!$

d)
$\binom{20}{1}$

e)
$\binom{20}{0}$

f)
$\binom{20}{2}$

g)
$\binom{20}{18}$

Chapter 8 · Original slide 16 · Binomial coefficients and Pascal's triangle

Number the top row 0 and the first entry 0. Entry k in row n is $\binom{n}{k}$. The top number gives the row. The bottom number gives the entry.

For large powers, such as $(x+3)^{20}$, calculate coefficients with the choose function rather than building all the rows.

Chapter 8 · Original slide 17 · Pascal's rule

This is useful enrichment for STEP. You are not required to know the proof for this course.

Chapter 8 · Original slide 18 · Exercise 8B – Y1 page 162

10 Given that $\binom{50}{13} = \dfrac{50!}{13!a!}$, write down the value of a.

11 Given that $\binom{35}{p} = \dfrac{35!}{p!18!}$, write down the value of p.

Chapter 8 · Original slide 19 · Exercise 8B – Y1 page 162

Challenge
a Work out ${}^{10}C_3$ and ${}^{10}C_7$.
b Work out ${}^{14}C_5$ and ${}^{14}C_9$.
c What do you notice about your answers to parts **a** and **b**?
d Prove that ${}^{n}C_r = {}^{n}C_{n-r}$.

Chapter 8 · Original slide 20 · Why binomial coefficients count the terms

The entries in row 5 are $1, 5, 10, 10, 5, 1$, or $\binom{5}{0}, \binom{5}{1}, \ldots, \binom{5}{5}$.

$(a+b)^5 = (a+b)(a+b)(a+b)(a+b)(a+b)$

Every term comes from choosing one term from each bracket. How many choices produce a^3b^2?

Chapter 8: The binomial expansion

Chapter 8 · Original slide 21 · The binomial expansion formula

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$$

For n a non-negative integer, each term is $\binom{n}{r} a^{n-r} b^r$. The two powers add up to n. Fractional and negative powers are studied in Year 2.

Example

Find the first four terms of $(3x+1)^{10}$, in ascending powers of x.

Chapter 8 · Original slide 22 · Test your understanding

First three terms

Find the first three terms of $(2 - \frac{1}{3}x)^7$, in ascending powers of x.

Chapter 8: The binomial expansion

Chapter 8 · Original slide 23 · Exercise 8C – Y1 page 164

5 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(2-x)^6$ and simplify each term.

7 Find the binomial expansion of $\left(x+\frac{1}{x}\right)^5$ giving each term in its simplest form.

6 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(3-2x)^5$ giving each term in its simplest form.

Chapter 8 · Original slide 24 · Extension

1

[AEA 2013 Q1a] In the binomial expansion of $\left(1+\frac{12n}{5}x\right)^n$ the coefficients of x^2 and x^3 are equal and non-zero.

Find the possible values of n.

Hint: Remember that $\binom{n}{2} = \frac{n(n-1)}{2!}$

Can you similarly simplify $\binom{n}{3}$ using $\binom{n}{r} = \frac{n!}{r!(n-r)!}$?

2

[STEP I 2010 Q5a] By considering the expansion of $(1+x)^n$, where n is a positive integer, or otherwise, show that:

$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{n} = 2^n$

reflection: This means that the sum of each row in Pascal's Triangle gives successive powers of 2. Safe!

Chapter 8: The binomial expansion

Chapter 8 · Original slide 25 · Selecting a general term

$T_{r+1} = \binom{n}{r} a^{n-r} b^r$

Expression	Power of x wanted	Required term
$(a+x)^{10}$	3	
$(2x-1)^{75}$	50	
$(3-x)^{12}$	7	
$(3x+4)^{16}$	3	

The powers of the two terms always add up to the power of the original bracket.

Chapter 8 · Original slide 26 · Finding a coefficient

The coefficient of x^4 in $(1+qx)^{10}$ is 3360. Find the possible values of q.

Chapter 8 · Original slide 27 · Comparing coefficients

Test your understanding

In $(1+ax)^{10}$, a is non-zero and the coefficient of x^3 is twice the coefficient of x^2. Find a.

Chapter 8 · Original slide 28 · Exercise 8D · Question 5

When $(1-2x)^p$ is expanded, the coefficient of x^2 is 40. Given that $p>0$, find (a) p, (b) the coefficient of x, and (c) the coefficient of x^3.

Chapter 8 · Original slide 29 · Exercise 8D – Y1 page 166-167

6 a Find the first three terms, in ascending powers of x, of the binomial expansion of $(5 + px)^{30}$, where p is a non-zero constant.
b Given that in this expansion the coefficient of x^2 is 29 times the coefficient of x work out the value of p.

Chapter 8 · Original slide 30 · Exercise 8D – Y1 page 166-167

7 a Find the first four terms, in ascending powers of x, of the binomial expansion of $(1 + qx)^{10}$, where q is a non-zero constant.
b Given that in the expansion of $(1 + qx)^{10}$ the coefficient of x^3 is 108 times the coefficient of x, work out the value of q.

Chapter 8 · Original slide 31 · Exercise 8D – Y1 page 166-167

8 a Find the first three terms, in ascending powers of x, of the binomial expansion of $(1 + px)^{11}$, where p is a constant.
b The first 3 terms in the same expansion are $1, 77x$ and qx^2, where q is a constant. Find the value of p and the value of q.

Chapter 8 · Original slide 32 · Exercise 8D – Y1 page 166-167

9 a Write down the first three terms, in ascending powers of x, of the binomial expansion of $(1 + px)^{15}$, where p is a non-zero constant.
b Given that, in the expansion of $(1 + px)^{15}$, the coefficient of x is $(-q)$ and the coefficient of x^2 is $5q$, find the value of p and the value of q.

Chapter 8 · Original slide 33 · Exercise 8D · Question 10

In $(1+x)^{30}$, the coefficients of x^9 and x^{10} are p and q. Find q/p.

Chapter 8 · Original slide 34 · Extension

1

[MAT 2014 1G] Let n be a positive integer. The coefficient of x^3y^5 in the expansion of $\left(1+xy+y^2\right)^n$ equals:

A) n

B) 2^n

C) $\binom{n}{3}\binom{n}{5}$

D) $4\binom{n}{4}$

E) $\binom{n}{8}$

2

[STEP I 2013 Q6] By considering the coefficient of x^r in the series for $(1+x)(1+x)^n$, or otherwise, obtain the following relation between binomial coefficients:

$\binom{n}{r}+\binom{n}{r-1}=\binom{n+1}{r}$

Chapter 8 · Original slide 35 · Estimating powers

(a) Find the first four terms of $(1+\frac{x}{4})^8$, in ascending powers of x, giving each term in its simplest form. (4 marks)
(b) Use the expansion to estimate 1.025^8 to four decimal places. (3 marks)

Chapter 8 · Original slide 36 · Test your understanding · Estimation

(a) Find the first four terms of $(1+\frac{x}{2})^{10}$, in ascending powers of x, in simplest form. (4 marks)
(b) Use the expansion to estimate 1.005^{10} to five decimal places. (3 marks)

Chapter 8 · Original slide 37 · Exercise 8E · Question 7

(a) Find the first three terms of $(3 - \frac{x}{5})^{10}$, in ascending powers of x, in simplest form. (4 marks)
(b) Explain how to use the expansion to estimate 2.98^{10}. (1 mark)

Chapter 8 · Original slide 38 · Exercise 8E · Question 8

(a) Find the first four terms of $(1 - 3x)^5$, in ascending powers of x, in simplest form. (4 marks)
(b) For small x, ignore x^2 and higher powers and show that $(1 + x)(1 - 3x)^5 \approx 1 - 14x$. (2 marks)

Chapter 8 · Original slide 39 · Exercise 8E – Y1 Page 168-169

9 A microchip company models the probability of having no faulty chips on a single production run as:

$$\mathrm{P}(\text{no fault}) = (1-p)^n,\ p < 0.001$$

where p is the probability of a single chip being faulty, and n being the total number of chips produced.

a State why the model is restricted to small values of p. (1 marks)

b Given that $n = 200$, find an approximate expression for $\mathrm{P}(\text{no fault})$ in the form $a + bp + cp^2$. (2 marks)

c The company wants to achieve a 92% likelihood of having no faulty chips on a production run of 200 chips. Use your answer to part **b** to suggest a maximum value of p for this to be the case. (4 marks)